MIL-P-55355A(ER)
3 March 1982
SUPERSEDING
MIL-P-55355(EL)
7 March 1973

MILITARY SPECIFICATION

PROJECTION SET, MOTION PICTURE, SOUND, 16MM

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of commercial, or modified commercial, motion picture projection set (see 6.4) and the following units thereof:

a. A portable 16 millimeter (mm) optical sound motion picture projector with integral amplifier and loudspeaker(s).
b. An external loudspeaker built into the removable cover of the integral equipment case.
c. A 1600 foot take-up reel.
d. A transit/storage case option (see 3.19).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) different issues than those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6730

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
T-T-871	- Twine, Fibrous Cotton, Wrapping.
W-C-596	- Connector, Cable Outlet, Electrical, Specific Purpose, Cable Connecting, General Grade, locking.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard; Corrugated, Triple-wall.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	- Strapping, Nonmetallic, (and Connectors).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation-packaging, Methods Of.
MIL-R-214	- Reel, Photographic Film, And Can Or Case, Photographic Film Storage (16mm and 35mm Film).
MIL-E-6060	- Envelope, Packaging, Water Vaporproof, Flexible.
MIL-P-11268	- Parts, Materials and Processes used in Electronic Equipment.
MIL-P-14232	- Parts, Equipment and Tools for Army Material, Packaging Of.
MIL-P-19644	- Plastic Molding Material (Polystyrene Foam, Expanded Bead).
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, For Packaging.

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging, and Packing Materials, Test Procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-150	- Photographic Lenses.
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1857(EL)	- Grounding, Bonding and Shielding Design Practices.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) different issues than those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE

PH22.8	- Projectable Image Area on 16mm Motion-Picture Film, Dimensions of.
PH22.24	- Transverse Cemented Splices on 16mm and 8mm Type R (Regular 8) Motion-Picture Film, Projection Type, Dimensions of.
PH22.42	- 16mm Sound-Focusing Test Films, Photographic Types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

PH22.43	- 16mm 3000-cycle Flutter Test Film, Photographic Type.
PH22.44	- 16mm Multifrequency Test Film, Photographic Type.
PH22.45	- 16mm 400-cycle Signal-Level Test Film, Photographic Type.
PH22.57	- 16mm Buzz-Track Test Film, Photographic Type.
S1.4	- Acoustics.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, New York 10018.)

SOCIETY OF MOTION PICTURE AND TELEVISION ENGINEERS (SMPTE)

16RT	- 16mm Registration Test Film.
16TG	- Travel Ghost Test Film.
P16PP (Jiffy)	- 16mm Projector Performance Test Film.
U16-JW-100	- Universal Jitter, Weave and Travel Ghost Test Film.

(Application for copies should be addressed to the Society of Motion Picture and Television Engineers, 862 Scarsdale Avenue, Scarsdale, New York 10583.)

UNDERWRITERS' LABORATORIES, INC.

UL-506	- Specialty Transformers.
UL-547	- Thermal Protection for Motors.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 42-15 Crescent, Long Island City, New York 11101.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Description. Projection Set, Motion Picture, Sound, 16mm covered by this specification includes a motion picture projector, a compatible external loudspeaker built into the removable cover of the integral equipment case and a 1600 foot take-up reel. The projector, which can be used with or without the external speaker, consists of a single-case unit with built-in amplifier and loudspeaker(s) and a zoom projection lens.

3.2 First article. When specified in the contract, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.3 Underwriters' Laboratories approval. The projection set shall conform with the standards of Underwriters' Laboratories, Inc., with regard to fire and casualty hazards. Contractor shall furnish statement of conformance from a competent source or Underwriters' Laboratories, Inc. (see 6.2).

3.4 Parts, materials and processes. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.4.1 Recycled, virgin and reclaimed materials. Without jeopardizing the intended use of equipment furnished to this specification, selection of materials will be governed by the following:

a. There is no exclusion to the use of recovered materials.
b. There is no requirement that equipment be manufactured from virgin materials.
c. Reclaimed materials are required to the maximum extent possible.

3.5 Mechanical requirements.

3.5.1 Carrying case. The projector shall be equipped with a carrying case of sturdy construction. The case may be an integral part of the component.

3.5.1.1 Carrying handle. The case shall be supplied with a carrying handle of adequate strength to withstand rough handling during transportation of the equipment.

3.5.1.2 Corners. The carrying case shall have no sharp corners or edges.

3.5.2 Storage of subcomponents.

3.5.2.1 Projector. The following items shall be stored in the projector:

a. Power cord.
b. The projection lens.
c. The 1600 foot reel.
d. External loudspeaker and its interconnecting cable (see 3.11.3.4).

3.5.3 Weight. The maximum weight of the projector including all items stored in it shall be 48 pounds. Paragraph 5.9.11.3.1.2 Labeling of MIL-STD-1472B shall be applicable.

3.6 Electrical requirements.

3.6.1 Input voltage and frequency. The projector shall be capable of operating over the range of 115 VAC $\pm$ 10 percent, 50 and 60 Hz $\pm$ 1 percent (see 4.25).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.2 Power cable. The cable used for the electrical connection between the source of electrical power and the projector shall be three conductor flexible cord, of a size approved by the Underwriters' Laboratories for the intended use and shall be at least 8 feet (2.44 meters) long. The plug of this cable shall comply with the last 3 sentences of paragraph 3.5 of W-C-596E, 11 Aug 78, which require a plug of a solid face, dead front, non-armored construction, with no insulating discs and no metallic cable clamps. The colors of the three conductors shall conform to single phase two-wire ac service requirements (1) of paragraph 3.2.8 of MIL-STD-1857(EL), 30 Jun 76. The green conductor shall be used as the grounding conductor between the equipment chassis and the ground of the power receptacle.

3.6.3 Grounding. Ground connections to shields, hinges, and other mechanical parts shall not be made to complete electrical circuits. The chassis or frame shall not serve as the common tie point of the neutral carrying wire. Ground connection to the chassis or frame shall be mechanically secured by soldering to a spotwelded terminal lug or to a portion of the chassis or frame that has been formed into a soldering lug, or by use of a terminal on the ground wire and then securing the terminal by a screw, nut, and a lockwasher.

3.6.4 Wiring. Stranded conductors, of suitable degree of flexibility, shall be used. All wire shall be of such cross section as to provide ample and safe current-carrying capacity and mechanical strength. The wiring layout shall be arranged to preclude exposure to detrimentally high local temperatures such as that encountered in the vicinity of the projection lamp. Where wiring passes through holes in metal, suitable insulation protection shall be provided.

3.6.5 Overload protection. Fuses or other suitable devices shall provide necessary overload protection. The overload protection shall have sufficient electrical rating to provide protection to the circuits.

3.6.6 Electromagnetic interference. The projection set shall comply with the emission requirements of Notice 4, MIL-STD-461A as specified in the following (see 4.44):

Method	Frequency range	Limits
RE02	.015 MHz to .150 MHz	120 dBuV/M/MHz
	.150 MHz to 400 MHz	100 dBuV/M/MHz
CE04B	.150 MHz to .500 MHz	114 dBuV/MHz
	.500 MHz to 100 MHz	88 dBuV/MHz

3.6.7 Thermal stability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.6.7.1 Motor running; heating. The motor shall be run continuously at maximum load until constant temperatures are attained. The maximum allowable winding temperature for class A insulation shall be no greater than 140°C (285°F), and for class B insulation, shall be no greater than 165°C (329°F) as stated in table 6.1, UL-547, 25 Oct 74, when tested in accordance with 4.25.1.

3.6.7.2 Transformer heating. The transformer, with the frame or enclosure grounded, shall be operated continuously at maximum rated primary load until constant temperatures are attained. The temperature rise at any point on the transformer frame or enclosure shall not be greater than 65C° (117F°), the temperature rise on class A insulation shall be not greater than 70C° (126F°), and the temperature rise on class B insulation shall be not greater than 95C° (171F°), as stated in 23.1 and 23.2, UL-506, 9 Apr 73. Other class insulation may be used providing the temperature rise at any point on the insulation does not exceed 95C° (171F°). Tests shall be performed in accordance with 4.25.1.

3.7 Projection requirements.

3.7.1 Film speed. The film shall travel at the standard sound speed of 24 frames per second ± 2.0 percent at 50 Hz ± 1 percent and 60 Hz ± 1 percent and 115 VAC ± 1 volt (see 4.6.1).

3.7.2 Sprocket feed. The sprocket shall be capable of feeding film having two adjacent damaged or broken perforations without causing further damage to the film (see 4.6.2).

3.7.3 Film reels. The projector shall include a 1600 foot (487.68 meters) type I steel reel in accordance with MIL-R-214. The projector shall operate satisfactorily with feed and take-up reels of all standard capacities from 400 to 2,000 feet (121.92 to 609.60 meters) (see 4.7).

3.7.4 Film rewinding mechanism. The film rewinding mechanism shall be power driven. It shall be capable of handling all lengths of film up to 2,000 feet (609.6 meters) without the necessity of removing reels from the projector or shifting or twisting belts and without allowing the film to contact any part of the film moving mechanism. The time required to rewind 2,000 feet (609.6 meters) of film, without damaging the film, shall be no more than 4.5 minutes. The lamp shall be off during the operation of the rewind mechanism (see 4.8).

3.7.5 Provision for framing. The projector shall have means of framing of the picture when the projector is in operation. The range of the framing mechanism shall extend at least 0.025 inches (0.635mm) above and below the standard position measured at the film and shall be free from creep when set. The means of framing shall not change the picture position on the screen by more than 1.0 percent of the picture height. The framing mechanism shall be easy to operate (see 4.9).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.6 Picture aperture. The effective picture aperture shall conform with ANSI PH22.8. The entire gate assembly shall be easily accessible for cleaning without the use of tools (see 4.10).

3.7.7 Projection lens. The projector shall be furnished with a zoom type 1.4 ±0.2 in (35.5 ±5.1mm) to 2.3 +0.3 or -0 in (58.4 + 7.6 or -0mm) f/1.6 or faster lens. If this lens is unmounted during transport, it shall have front and rear lens covers; if it is mounted during transport, it shall include at least a front lens cover. The lens shall conform with the requirements for type VII lenses of MIL-STD-150. The projector shall be capable of using anamorphic lenses and a 4 inch (101.6mm) focal length lens.

3.7.7.1 Lens interchangeability. The projector shall accommodate interchangeable lenses of the effective apertures and focal lengths stated in 4.11.

3.7.7.2 Mounting. The method of mounting shall assure that each lens is centered on the optical axis and is maintained with its axis perpendicular to the film within limits to satisfy the other requirements of this specification.

3.7.7.3 Focusing mechanism. The projector focusing mechanism shall have both coars (coars positioning will be acceptable in lieu of a coarse adjustment) and fine adjustments. The fine adjustment shall be easily operable while the projector is in use. The mechanism shall be free from back-lash and have sufficient friction so that it will not move or creep during operation.

3.7.7.4 Image quality of projection lens. The image produced by the projection lens shall be visually free from haze due to spherical and chromatic aberration, internal barrel reflections, surface and glass defects when viewed from a distance twice the width of the projected image.

3.7.7.5 Coatings. Coatings or blackenings used to deaden reflections shall be free from dusting or flaking resulting from exposure to any tests specified in this specification. Aluminum barrels, if used, shall be anodized.

3.7.7.6 Resolution. The minimum resolving power determined by bench test for the zoom lens when tested as specified in 4.43 shall be 65 lines per millimeter AWAR, with at least 100 lines per millimeter on axis and not less than 40 lines per millimeter off axis.

3.7.8 Tilt device. Means for tilting the projection axis from horizontal to 10 degrees above the horizontal shall be provided. When set, the adjustment shall remain fixed and shall not creep when the projector is operating (see 4.26).

3.7.9 Uniformity of screen illumination. Illumination on the screen shall be free from objectionable bands or patches differing in color or brightness from the adjacent parts of the screen. The average illumination at the four corners on the screen shall be not less than 60 percent of the illumination at the center. The illumination at any corner shall be not less than 50 percent of the illumination at the center of the screen. Measurements shall be made as specified in 4.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.10 Light output. The projector shall provide a projected light output of at least 350 lumens on a screen located so that the area lighted is at least 40 inches (1016mm) wide. The projection lamp of the projector shall have a rated life of at least 25 hours (see 4.13).

3.7.11 Flicker. There shall be a minimum of two interruptions of light for each frame of film. The light intervals shall be equal in duration and the dark intervals shall be equal in duration (see 4.27).

3.7.12 Picture unsteadiness.

3.7.12.1 Vertical unsteadiness. Vertical unsteadiness, having a period shorter than 1 second, shall not exceed 0.2 percent of the picture width in new projectors and shall not exceed 0.25 percent in projectors which have been operated for a period of 1000 hours (see 4.14).

3.7.12.2 Horizontal unsteadiness. Horizontal unsteadiness having any period whatsoever shall not exceed 0.2 percent of the picture width, either in new projectors or in projectors which have operated for a period of 1000 hours (see 4.14).

3.7.13 Freedom from travel ghost. No objectionable travel-ghost shall be visible with the framing mechanism in any position when the projector is tested with SMPTE 16-TG test film (see 4.15). Since SMPTE no longer supplies the 16-TG test film, their U16-JW-100 test film may be used for this test. No travel ghost exceeding one-half block shall be visible if SMPTE 16 RT test film is utilized.

3.7.14 Location of film in image plan of lens. The resolution on the four corner rosettes of SMPTE 16RT test film shall show not more than two rings spread, when tested per 4.16.

3.7.15 Film life. Newly processed and lubricated sound and picture film, in good condition and having a shrinkage not greater than 0.3 percent shall exhibit no visible evidence of damage in the projected picture, and no audible evidence of damage to the sound track in the reproduced sound after 1,000 passages through the projector with the lamp energized. Inspection of the film with the unaided eye shall reveal no evidence of damage to the film at the sprocket holes. Five well-made splices, in accordance with ANSI PH22.24 and made with new cement, equally spaced on a 12 foot (3.65 meter) loop, shall perform not less than 500 passages through the projector (see 4.34).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.7.16 Temperature rise of projector housing. Continuous operation of the projector at rated voltage, with the lamp energized, for a period of two hours shall not produce a temperature rise of any part of the projector, exclusive of the top of the lamp house cover and blower exhaust port, in excess of 28C° (50.4F°) above the ambient when tested as specified in 4.35.

3.7.17 Adequacy of lamp house ventilation. Continuous operation of the projector at rated voltage for a period of 10 hours, with the light source energized, and the ambient temperature at 33°C (91.4°F) shall cause no bulging of the lamp envelope or other damage to the projector due to excessive heating (see 4.36).

3.7.18 Noise of projector mechanism. The noise of the projector when operating with newly processed sound and picture film shall be not greater than 63 decibels (db) above a reference sound pressure level of 20 micro-newtons (2 x 10^{-4} microbar) as specified in ANSI S1.4 when measured in accordance with 4.37. The immediate vicinity of the air exhaust port shall be excepted from this requirement.

3.8 Sound requirements.

3.8.1 Soundhead.

3.8.1.1 Freedom from flutter. The total root mean square (r.m.s.) flutter shall be not greater than 0.30 percent when using a 3,000 Hz flutter test film in accordance with ANSI PH22.43, such as SMPTE P16FL. Since SMPTE now makes this film at a frequency of 3150Hz, this frequency is also acceptable. Wow peaks (random excursions) can be 0.6 percent maximum (see 4.17).

3.8.1.2 Scanning beam.

3.8.1.2.1 Dimensions. The scanning light beam, when properly focused upon the film being reproduced, shall have dimensions which will permit satisfactory performance of the amplifier.

3.8.1.2.2 Distance from scanning beam to picture. The film path shall be designed so that, when properly threaded, the corresponding sound track signal is 26 ± 0.5 frames ahead of the picture frame (see 4.23). If an automatic loop setter is used, the corresponding sound track signal may be 26(+2 or -1) frames ahead.

3.8.1.2.3 Lateral positioning of the film in the scanning light beam. The projector shall be adjusted so that the side tone from the buzz track test film complying with ANSI PH22.57, such as SMPTE P16BT, shall be at least 40db below the established rated power output referenced level (see 4.18).

3.8.1.2.4 Provision for focus of scanning beam. No adjustment shall be necessary for focusing the scanning light beam on either surface of the film. The focal setting shall provide maximum output from a 7,000 Hz focusing test film in accordance with ANSI PH22.42, such as SMPTE P16SFA, when it is run through the projector (see 4.19).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

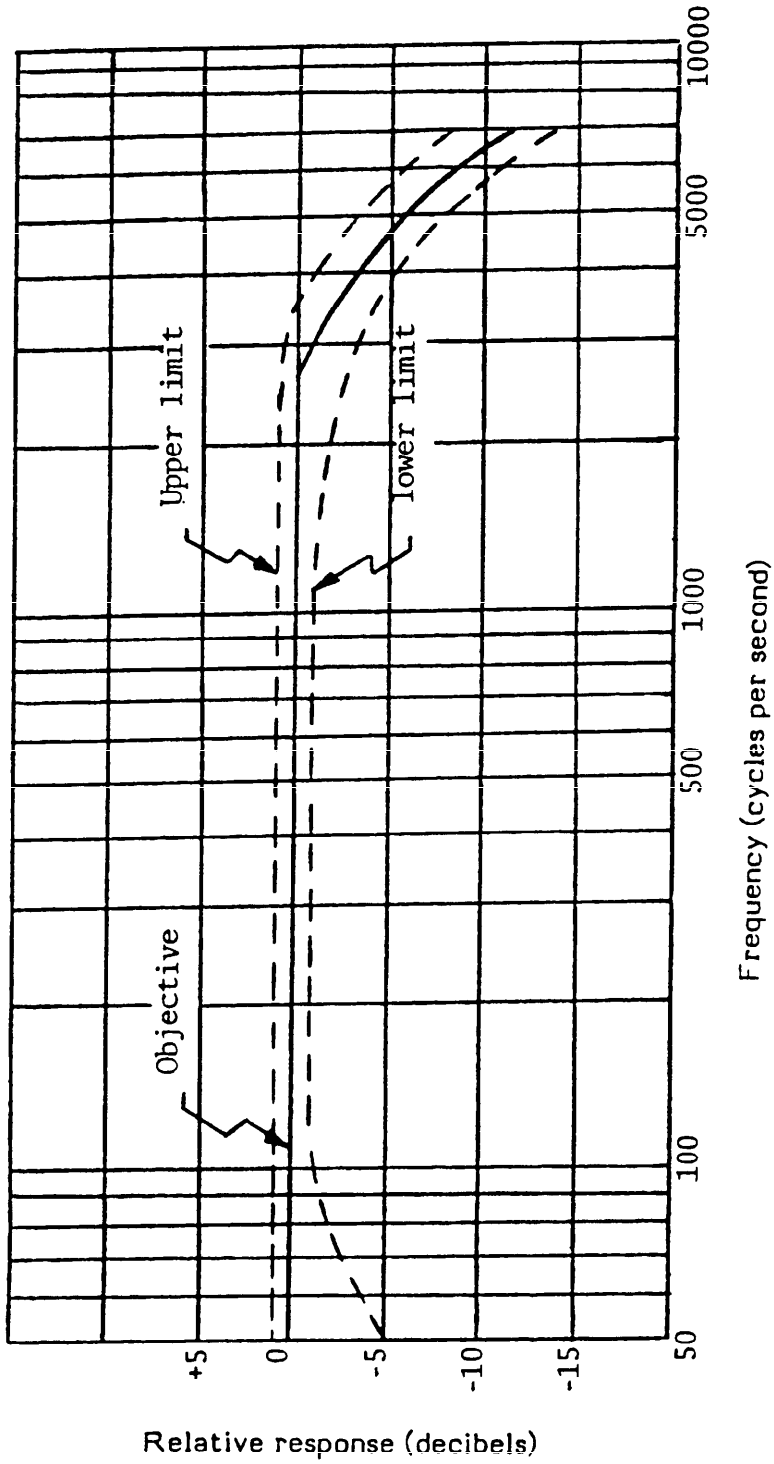


FIGURE 1. Amplifier frequency response characteristic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.8.1.2.5 Stray light. The sound-track sensor shall be adequately protected against stray light.

3.8.1.2.6 Sound-track sensor adjustment. The sound-track sensor adjustment shall be simple to operate, or shall be a prefocused assembly which will permit an accurate replacement of the sensor and which, after adjustment, shall remain fixed for maximum output during operation.

3.9 Amplifier requirements.

3.9.1 General. The amplifier shall be installed within the projector carrying case. The amplifier shall be solid state. Miniaturized components shall be used where possible.

3.9.2 Serviceability. The amplifier assembly shall be readily removable for service, repair and/or replacement.

3.9.3 Power output and distortion. The amplifier shall deliver an output of at least 7.5 watts, with a total harmonic content not exceeding 2.0 percent at any frequency from 50 to 7000 Hz (see 4.20).

3.9.4 Amplifier gain. The amplifier gain shall be sufficient to develop an output of 7.5 watts from a 400 Hz test film in accordance with ANSI PH22.45, such as SMPTE P16SL, when the tone control is set for flattest response and the volume control is set for an attenuation of 15db (see 4.21).

3.9.5 Operating controls. Operating controls shall include a loudspeaker selector switch or a jack outlet which will automatically switch from internal to external speaker, a power switch, a volume control, and a tone control. The volume control shall be capable of completely cutting out each speaker. The speaker selector shall be connected to cut-in and cut-out the internal and external loudspeaker. Each position shall be properly marked.

3.9.5.1 Volume control. The volume control of the amplifier shall provide a working range of attenuation of not less than 40 db within which the relation of rotation in db is approximately linear. A reference scale for the convenience of the operator shall be provided. Movement of the volume control shall produce no audible noise in the output. The volume control shall be in the amplifier circuit and shall not control the exciter lamp voltage (see 4.28).

3.9.5.2 Tone control. One tone control shall be provided affecting substantally the frequency range above 500 Hz of the amplifier (see 4.29).

3.9.6 Amplifier frequency response characteristic. When measured with ANSI PH22.44 film such as P16MF, the amplifier shall have a frequency response complying with figure 1 (see 4.30).

3.9.7 External loudspeaker receptacle. The projector amplifier shall be provided with an external loudspeaker receptacle (see 4.22).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.10 Dynamic range. The maximum overall electrical noise level of the equipment (projector and amplifier) under normal operation and running without film shall be at least 40 db below the rated output of the amplifier, when measured electrically at the loudspeaker terminals (see 4.31).

3.11 Loudspeakers.

3.11.1 Construction. The loudspeakers shall be of the permanent magnet moving voice coil type.

3.11.2 Diaphragm materials. The diaphragms shall be of such materials as will not suffer permanent damage or degradation of performance when the loudspeakers are subjected to environmental tests specified herein.

3.11.3 External loudspeaker.

3.11.3.1 Voice coil impedance. The impedance of the voice coil shall match that of the projector amplifier output.

3.11.3.2 Power rating. The loudspeaker shall have a power rating of at least 15 watts.

3.11.3.3 Size. If the external loudspeaker contains one speaker, it shall be at least of 12 inches (30.48cm) diameter. If it contains more than one speaker, it is exempt from this requirement, but shall meet all other requirements of this specification.

3.11.3.4 Interconnecting cable. An interconnecting cable at least 50 feet (15.24m) but not more than 65 feet (19.81m) in length shall be supplied to connect the loudspeaker to the projector.

3.11.4 Power handling capability. The power handling capabilities of both the internal and external loudspeakers shall be such that one input of representative program material to each of the two loudspeakers will not generate distortion, spurious noises or rattles which impair the quality of the music or speech reproduced when the input, as measured on an oscilloscope, has peaks as follows:

a. Integral loudspeaker, 10 watts

b. External loudspeaker, equal in watts to the power output rating of the amplifier, but not less than 10 watts (see 4.32).

3.11.5 Dustproofing. Complete dustproofing shall be provided in the loudspeakers to prevent dust and metal particles from interfering with their performance.

3.12 Service conditions. The equipment shall meet the following service conditions:

3.12.1 Temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.12.1.1 Low. During testing as specified in 4.38.1, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
4.23	Operation at +40°F (4.4°C)	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.12.1.2 High. During testing as specified in 4.38.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
4.23	Operation at 122°F (50°C)	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.12.2 Humidity (moisture-resistance). Each component of the equipment shall meet full specification performance for the following measurements both during and after testing as specified in 4.39:

Paragraph	Measurement
4.23	Operation

13.12.3 Altitude (elevation). Each component of the equipment shall meet full specification performance for the following measurement both during and after testing as specified in 4.40:

Paragraph	Measurement
4.23	Operation

3.12.4 Vibration. The projector, in its self-contained carrying case, and the extension speaker shall be vibrated in the frequency range of 5-55-5 Hz for 30 minutes in each of the three (3) mutually perpendicular planes in accordance with 4.41.

3.13 Durability. The projector shall be capable of operating with film for 1000 hours without loss of efficiency and without appreciable wear and the amplifier shall show no change in its characteristics when tested in accordance with 4.42. It shall not be necessary to replace take-up or rewind belts during this test.

3.14 Interchangeability. All parts, components, and subassemblies having the same manufacturer's part number shall be physically, dimensionally, and functionally interchangeable (see 4.33).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

3.15 Finish. All plating and finishes shall be in accordance with best commercial practices, and shall enable the equipment to meet the environmental conditions specified herein.

3.16 Marking. The equipment shall be provided with a nameplate in accordance with MIL-STD-130.

3.16.1 Threading diagram. The projector shall contain a film threading diagram placed in a conspicuous location.

3.16.2 Lens marking. The lens shall be marked for effective focal length and maximum relative aperture in accordance with commercial practice.

3.16.3 Visibility. Wherever practicable, identification markings shall be visible with minimum disassembly of the equipment.

3.16.4 Serial numbers. The equipment shall have a serial number.

3.17 Operation. The audio characteristics of the equipment shall be such that operation with SMPTE P16PP (Jiffy) projector performance test film results in no distortion, spurious noises or rattles which impair the quality of the sound reproduced (see 4.23).

3.18 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454, and the applicable portions of the following paragraphs:

3.6.2 Power cable
3.6.3 Grounding
3.6.4 Wiring
3.6.5 Overload protection
3.20 Safety

3.19 Transit-storage case option. When the projection set is intended for tactical field use, a case shall be provided for transit and storage of the equipment. The case shall be rigid, reuseable, and rainproof, manufactured from ferrous or non-ferrous metals and non-metallic materials such as high density linear polyethylene or other compounded, non-metallic materials.

3.20 Safety. The equipment shall meet the requirements of MIL-STD-454, requirement 1 (see 4.48).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.46).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor (see 6.2c).

4.3.1 First article units. When required by contract, the contractor shall furnish first article units of the complete projection set described herein.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

Inspection	Reqt Para	Insp Para	Unit 1	Unit 2	Unit 3
Inspections covered by subsidiary documents	3.7.7 3.18	4.4 4.45	Inspections to be performed on all units.		
Group A	See table II		Inspections to be performed on all units.		
Group B	See table III		Inspections to be performed on all units.		
Group C	See table IV		Group I	Group II	Group III

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government, except that the operational test, 4.23, shall be last.

4.5.2 Group B inspection. Group B inspection shall be performed on inspected lots that have passed group A inspection. This inspection shall conform to table III of the specification and to the inspection levels contained in 4.5.2.1 (reference table I of MIL-STD-105).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual & Mechanical	3.18	4.45
Carrying case	3.5.1-3.5.1.2	4.45
Projector	3.7.7.2, 3.7.7.3 3.7.7.5, 3.8.1.2.6	4.45
Amplifier	3.9.1, 3.9.2, 3.9.5, 3.9.7	4.45
Loudspeakers	3.11.1, 3.11.2, 3.11.3.3, 3.11.3.4, 3.11.5	4.45
Performance		
Film speed	3.7.1	4.6.1
Film reels	3.7.3	4.7
Film rewinding	3.7.4	4.8
Framing	3.7.5	4.9
Picture aperture	3.7.6	4.10
Lens interchangeability	3.7.7, 3.7.7.1	4.11
Uniformity of screen illumination	3.7.9	4.12
Light output	3.7.10	4.13
Vertical & Horizontal unsteadiness	3.7.12.1 3.7.12.2	4.14
Travel ghost	3.7.13	4.15
Location of film in image plane of lens	3.7.14	4.16
Flutter	3.8.1.1	4.17
Lateral positioning of film	3.8.1.2.3	4.18
Focus of scanning beam	3.8.1.2.4 3.8.1.2.5	4.19
Amplifier power output	3.9.3	4.20
Amplifier gain	3.9.4	4.21
External loudspeaker receptacle	3.9.7	4.22
Operational	3.7.7.4 3.8.1.2.2 3.9.5, 3.17	4.23

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group B sampling plans. The group B sampling plans, for the AQL's listed in table III, shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL percent defective (see 4.5.2.1)
Weight	3.5.3	4.24	6.5%
Input voltage & frequency	3.6.1	4.25	6.5%
Sprocket feed	3.7.2	4.6.2	6.5%
Tilt	3.7.8	4.26	6.5%
Resolution	3.7.7.6	4.43	6.5%
Flicker	3.7.11	4.27	4%
Volume control	3.9.5.1	4.28	6.5%
Tone control	3.9.5.2	4.29	6.5%
Amplifier frequency response	3.9.6	4.30	4%
Dynamic range	3.10	4.31	4%
Loudspeaker power handling capacity	3.11.4	4.32	6.5%
Interchangeability	3.14	4.33	6.5%

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

4.5.2.3 Rejected lots. If an inspected lot is rejected, the manufacturer may rework it to correct the defects or to screen out the defective units and resubmit the lot for inspection. Resubmitted lots shall be inspected using tightened inspection procedures of MIL-STD-105. Resubmitted lots however, shall not be tendered for acceptance unless the former cause of rejection or requirement of correction is disclosed in writing. Such lots shall be kept separate from new lots and be clearly identified as resubmitted/reinspected lots.

4.5.3 Group C inspection. Group C inspection shall be performed on actual units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Group C inspection.

Introduction	Requirement paragraph	Test method paragraph
Group I		
Durability	3.13	4.42
Group II		
Film life	3.7.15	4.34
Low temperature	3.12.1.1	4.38.1
High temperature	3.12.1.2	4.38.2
Temperature rise	3.7.16	4.35
Adequacy of ventilation	3.7.17	4.36
Altitude (elevation)	3.12.3	4.40
Noise of projector mechanism	3.7.18	4.37
Humidity (moisture-resistance)	3.12.2	4.39
Group III		
Electromagnetic interference	3.6.6	4.44
Vibration	3.12.4	4.41

4.5.3.1 Sampling for inspection. One sample unit shall be selected at random for each group of inspections in table IV from every 100 and fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot (see 4.5.3).

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in accordance with order shown in table IV.

4.5.3.3 Group C failures. Action required relative to group C failures shall be as specified in the contract. Any failure of the sample tested shall be cause for rejection of the entire lot or batch represented by the sample.

4.5.4 Reinspection of group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract provided all damage to the completed units and their components is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Film transport and re-engaging test.

4.6.1 Film speed. A suitable length of film containing a known number of frames shall be timed by a stopwatch in its passage through the projector, operating at 50 Hz ± 1 percent and 60 Hz ± 1 percent and 115 VAC ± 0.1 volt. The corresponding film speed shall then be calculated to determine compliance with 3.7.1. Other means of measuring film speed will be considered acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.6.2 Film feed. A suitable length of film having two adjacent perforations removed by cutting the film as shown in figure 2 shall be fed through the projector to determine compliance with 3.7.2.

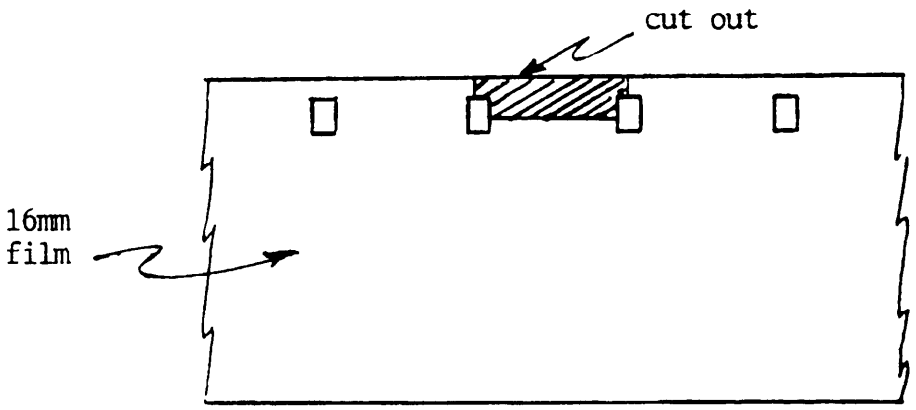


FIGURE 2. Film feed test.

4.7 Film reel combinations. The equipment shall be tested to determine whether it will operate satisfactorily with various combinations of 400-, 800-, 1,600-, and 2,000-foot capacity film reels to determine compliance with 3.7.3.

4.8 Rewinding time. The time required to rewind a 2,000-foot reel of film shall be measured to determine compliance with 3.7.4.

4.9 Framing. The range of the framing mechanism and its effect on the picture position on the screen when it is operated shall be measured to determine compliance with 3.7.5. Any creepage shall be cause for rejection.

4.10 Picture aperture. The effective picture aperture shall be measured in accordance with ANSI PH22.8 to determine compliance with 3.7.6.

4.11 Lens interchangeability. The projector shall be checked to determine whether it will satisfactorily operate with the f/1.6, zoom type, 1.4 to 2.3 inches and a 4 inch f/2.8, or faster lenses to determine compliance with 3.7.7.1.

4.12 Uniformity of screen illumination. The projector shall be focused on the film plane and the illumination shall be measured on a screen not less than 40 inches wide. Measurement of illumination shall be made at the center and at four points in the corners located one-twentieth of the screen width from the top or bottom edge and at the same distance from the side edges of the screen. The average illumination at the four corner points shall be computed, and any objectional bands or patches differing in color or brightness from adjacent parts of the screen shall be noted to determine compliance with 3.7.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.13 Light output. This test shall be performed following the test for uniformity of screen illumination to determine compliance with 3.7.10. There shall be no adjustments made to the uniformity of screen illumnation. The light output shall be measured with the shutter running, but with no film in the gate. Measurements shall be made in the plane of a screen perpendicular to the optical axis of the projector and located at a distance which will provide a projected image of the picture aperture at least 40 inches wide.

4.13.1 Method. The illuminated area of the screen shall be divided into 12 equal areas. The direct illumination at the center of each area shall be measured by means of a photoelectric light meter corrected by suitable color filters to conform within a good approximation to the visibility curve of the eye. The arithmetical average of the 12 illumination measurements in foot-candles shall be multiplied by the area of the screen in square feet to obtain the light output in lumens. The projector shall be operated at 115 VAC $\pm$ 0.1 volt, 50 and 60 Hz $\pm$ 0.1 Hz with a lens that has a transmittance equivalent to a 2 inch focal length lens having an aperture of f/1.6.

4.14 Picture unsteadiness test. The vertical and horizontal picture unsteadiness shall be tested using SMPTE test film 16 RT to determine compliance with 3.7.12.

4.15 Travel ghost. The presence or absence of travel ghost shall be determined using a test film in accordance with SMPTE 16-TG (or U16-JW-100) or 16RT to determine compliance with 3.7.13.

4.16 Location of film in image plane of lens. The sharpness of focus of the four corner targets visible, when SMPTE registration test film 16RT is projected to a 40 inch wide image and viewed at a distance twice the width of the screen, shall be examined to determine compliance with 3.7.14.

4.17 Flutter. The total root-mean-square (rms) flutter shall be measured by the bridge method using a 3,000 Hz signal from a test film in accordance with ANSI PH22.43 to determine compliance with 3.8.1.1. The measurements shall be made as soon as practicable after 6 feet of film have passed through the projector.

4.18 Lateral positioning. The lateral beam adjustment shall be checked using a buzz-track test film in accordance with ANSI PH22.57 to determine compliance with 3.8.1.2.3.

4.19 Focus of scanning beam. The focus of the scanning beam shall be tested for maximum output at 7,000 Hz using a test film in accordance with ANSI PH22.42 to determine compliance with 3.8.1.2.4 and 3.8.1.2.5. This test shall be conducted using a loop of film consistng of two equal lengths of the test film so spliced that the sound track of one length is toward the sound optics and the other away from the sound optics. The scanning beam shall be focused so that the outputs of the two films do not vary by more than 3 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.20 Amplifier power output. The total harmonic contents of the amplified output at frequencies of 50, 100, 400, 1000, 2000, 3000, 5000 and 7000 Hz shall be measured using a signal generator having a harmonic of not more than 0.3 percent. The level of the input signal shall be such that 7.5 watts output is obtained at substantially the same gain setting as is required for 7.5 watt output when reproducing a 400 Hz test film in accordance with ANSI PH22.45. The amplifier output shall be connected to a non-inductive resistive load. Power output and distortion shall be as specified in 3.9.3. During this test, the projector shall be operating with light source energized.

4.21 Amplifier gain. With the tone control set at normal response and the volume control at an attenuation of 15 db, a 400 Hz test film in accordance with ANSI PH22.45 shall be passed through the projector and the output power measured into a non-inductive resistive load to determine compliance with 3.9.4.

4.22 Loudspeaker output receptacles. The external loudspeaker output receptacle shall be tested for continuity to determine compliance with 3.9.7.

4.23 Operational tests.

4.23.1 Projector. The projector shall be checked for final adjustment using SMPTE P16PP (Jiffy) projector performance test film to determine compliance with 3.7.7.4, 3.8.1.2.2, 3.9.5 and 3.17.

4.23.2 External loudspeaker. The loudspeaker shall be checked for operational suitability using the audio portion of SMPTE P16PP (Jiffy) projector performance test film to determine compliance with 3.17.

4.24 Weight. The projector including all items stored in it shall be weighed to determine compliance with 3.5.3.

4.25 Input voltage and frequency. The projector, threaded with a 2,000 foot reel of film, shall be successively started and stopped 20 times under the following four line input conditions to determine compliance with 3.6.1:

a. 103.5 (+1 or -0) volts at 50 Hz $\pm$1%

b. 103.5 (+1 or -0) volts at 60 Hz $\pm$1%

c. 126.5 (+0 or -1) volts at 50 Hz $\pm$1%

d. 126.5 (+0 or -1) volts at 60 Hz $\pm$1%

4.25.1 Thermal stability. During the following tests, the projector reels shall be weighed to simulate a 2000 foot reel of film, and the audio circuit energized with output adjusted to 8 watts across an 8 ohm non-inductive resistive load. The projector lamp shall be at normal setting, and input voltage adjusted to 126.5 (+0 or -1) volts, 50 Hz $\pm$ 1 percent. The projector shall be continuously operated until temperatures stabilize. At the conclusion of tests, the projector shall meet the operational requirements of 4.23.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

a. Motor running; heating. Maximum allowable winding temperatures measured by means of thermocouples shall meet the requirements of 3.6.7.1.

b. Transformer heating. Maximum winding temperatures obtained shall meet the requirements of 3.6.7.2.

4.26 Tilt device. The tilt device shall be checked for compliance with 3.7.8.

4.27 Flicker. The number of interruptions of the light for each frame of film shall be determined. The duration of the light intervals shall be measured, and the duration of the dark intervals shall be measured to determine compliance with 3.7.11.

4.28 Volume control. The volume control attenuation in decibels shall be ascertained at a sufficient number of points to plot a curve of attenuation in decibels as a function of degrees rotation and it shall be noted whether rotation of the volume control produces audible noise in the output to determine compliance with 3.9.5.1.

4.29 Tone control. The response frequency curves of the amplifier shall be ascertained with the tone control in various positions using a constant voltage variable-frequency input to determine compliance with 3.9.5.2.

4.30 Amplifier frequency response. Use ANSI PH22.45 (400-cycle) test film and set gain for maximum rated output into a resistive load. The projector shall then be threaded with ANSI PH22.44 multifrequency film. The frequency response shall then be checked from 50 to 7000 Hz to determine conformance with 3.9.6.

4.31 Dynamic range. The equipment (projector and amplifier) shall be operated without film but with the motor running. The volume control shall be set at that position corresponding to a gain of 15 db in excess of that required to produce rated output using a 400 Hz test film in accordance with ANSI PH22.45. A resistive load shall be connected to the amplifier output terminals. The total electrical noise level at the amplifier output terminals shall be measured and compared with the rated 400 Hz output level at 5 percent distortion to determine compliance with 3.10.

4.32 Loudspeaker power handling capacity. The loudspeakers shall be tested for conformance with 3.11.4.

4.33 Interchangeability. Subassemblies shall be interchanged or substituted, any one for another, without modification or selection and shall provide the same physical and functional characteristics required of the original subassemblies and will require no modification of the projector.

4.34 Film life. A 12-foot loop of newly processed and lubricated sound and picture film in good condition, having a shrinkage not greater than 0.3 percent shall be threaded through all parts of the projector mechanism that touch the film in normal operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

The loop shall be supported in the return path in as near a normal position as possible by free-running idler rollers of such construction as to have a negligible tendency to damage the film. The loops shall be under a tension of not less than 2 nor more than 8 ounces. The loops shall be passed through the projector mechanism, with the projection lamp burning, for 1,000 times and the film examined for visible damage by projection and for audible damage by listening. The perforations shall also be examined for damage. The splices shall be replaced as may be necessary during the test. The life of 5 splices shall be noted, and the average number of passages of the 5 splices shall be computed to determine compliance with 3.7.15.

4.35 Temperature rise of projection housing. The projector, exciter lamp and all other sources of heat shall be operated continuously at line voltage of 115 volts and at an ambient temperature of 33° (+3 or -0)^{o}C. The temperature rise at the hottest spot of the projector housing, exclusive of the top two inches of the lamphouse, shall be determined by the thermocouple method when 2 hours have elapsed, to determine conformance with 3.7.16.

4.36 Adequacy of ventilation. The projector and all its sources of heat shall be operated at 115 volts for a period of 10 hours at an ambient temperature of 33° (+3 or -0)^{o}C. The projection lamp and projector shall then be examined for damage caused by excessive heating to determine compliance with 3.7.17.

4.37 Noise of projector mechanism. The projector shall be mounted in free field conditions on a small stand such that it can be turned on a vertical axis through the picture aperture. The microphone of a sound-level meter, in accordance with Type 2 (General Purpose) of ANSI S1.4, shall be placed 3 feet from this vertical axis and in a horizontal plane passing through the picture aperture. Readings of the sound level shall be taken at 45 degree intervals, beginning with a reference position in front of the projector and on the longitudinal axis of the picture aperture, while the projector is operating normally with film and with the meter at the A weighting setting, to determine compliance with 3.7.18.

4.38 Temperature test.

4.38.1 Low temperature. Each component of the equipment in its carrying case shall be tested in accordance with method 502, procedure 1, of MIL-STD-810. The storage temperature shall be -40°F (-40°C) for 12 hours, and the operating temperature shall be $+40^{o}$F (4.4^{o}C). Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.12.1.1.

4.38.2 High temperature. Each component of the equipment in its carrying case shall be tested in accordance with method 501, procedure II, of MIL-STD-810 for 2 cycles, omitting step 6. The operating temperature shall be 122°F (50°C). Prior to testing the equipment shall meet full specification performance for the measurement specified in 3.12.1.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the currentversion before use.

4.39 Humidity (moisture-resistance) test. Each component of the equipment in its carrying case shall be tested in accordance with method 507, procedure I of MIL-STD-810 with temperature at 95°F (35°C) and approximately 85 percent humidity at step 2 for 3 cycles. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.12.2. Measurements shall be made at the end of each of the three cycles with elimination of step 6 of procedure I. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.40 Altitude (elevation) test. Each component of the equipment in its carrying case shall be tested in accordance with method 500, procedure I, of MIL-STD-810, except that the internal pressure at step 2 should be 5.56 inches of Hg (equivalent to 40,000 feet altitude), omitting step 3 to determine compliance with 3.12.3.

4.41 Vibration test. The test items shall be tested in accordance with method 514.2, procedure X, curve AW, MIL-STD-810, for compliance with 3.12.4, except that the frequency range shall be 5-55 Hz and the sweep rate shall be 5-55-5 Hz in 7.5 minutes.

4.42 Durability test. The projector, with amplifier energized and 8 watts into a resistive load, shall be operated for a period of 1,000 hours with the light source energized. The projection lamp may be operated at 90 percent of its rated voltage in order to conserve its life. Lamps shall be replaced whenever they burn out. At the conclusion of the 1,000 hour run the equipment shall be required to meet the performance tests of group A inspection. At the conclusion of the performance tests, the projector shall be disassembled, and all parts examined for excessive wear (see 3.13). The average life of the projection lamps shall be noted in order to ascertain whether improper design has caused undue shortening of lamp life. This test shall not be interrupted except for the following reasons:

a. Replacement of bulbs.
b. Replacement of film.
c. Periodic oiling every 500 hours, if required.
d. Conformity with the testing laboratory working hours.
e. Damage to the projector.
f. Damage to the film.

4.42.1 Film loop. It is recommended that the film used in this test be in the form of a loop of such length that it does not require replacement at short intervals. Weights shall be used to simulate actual film tension and reel arm loading (2,000 foot reel loaded).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

4.42.2 Rewind mechanism life test. Upon completion of the durability run, the rewind mechansim shall be placed under the test using weights and tension to simulate the actual rewinding of a 2,000 foot reel of film. This test shall continue until it has been determined that the rewind mechanism in actual operation would have rewound 1,000,000 feet of film. This is to determine the adequacy of the rewind mechanism.

4.43 Resolution. The lens shall be tested in accordance with method 15 of MIL-STD-150 to determine compliance with 3.7.7.6.

4.44 Suppression. The projection set shall be tested for compliance with the requirements of 3.6.6. Test setups and procedures shall comply with the measurement techniques of Notice 3, MIL-STD-462.

4.45 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252 and table V.

TABLE V. Classification of visual and mechanical defects.

Classification	Defect
Major	Those defects, as listed in MIL-STD-150, which exceed the conditions governing beauty defects. Imperfections in the mounting ring or metal edging of the lens, which when the lens is secured in the projector could cause unequal strain or distortion of the lens. Supply or takeup mechanism binds, scrapes, or sticks when operated. Rate of film takeup too slow for projector speed. Idlers do not remain in position. Projector cinches film on takeup or supply reel. Fan does not operate. Motor does not operate. Focusing mechanism binds, scrapes, or sticks when operated. Projector tears film perforations. Projector scratches film during operation. Reels do not fit on projector spindle Projector does not retain film loop during operation.
Minor	Controls bind, scrape, or stick, when operated. Rewinder binds or scrapes while winding film. Idlers bind or scrape. Leveling adjustments bind, scrape, or stick when operated. Those defects, listed in MIL-STD-150, classified as beauty defects.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

4.46 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.47 First article inspection of packaging. When unit pack design validation in accordance with MIL-P-116 is required, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

4.23 Operational test.

4.48 Safety. In conformance with 3.20, safety caution marking shall comply with paragraph 9.2a of MIL-STD-454F.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Projection Set, Motion Picture Sound, shall be cleaned in accordance with process C-1 of MIL-P-116. Projection lens assembly shall be cleaned in accordance with MIL-P-14232.

5.1.1.2 Drying. Each Projection Set, Motion Picture Sound, shall be dried in accordance with the applicable procedure of MIL-P-116. Projection lens assembly shall be dried in accordance with MIL-P-14232.

5.1.1.3 Preservative application. Preservatives and their application shall be in accordance with MIL-P-116.

5.1.1.4 Prepackaging. The external speaker cable assembly and projector power cable assembly shall be individually wound or coiled as applicable, stowed in their designated locations and secured by retainer(s) provided or by twine conforming to T-T-871, type II. Stow and secure the remaining components of the projector in their designated locations. The projection lens assembly shall be covered with front and rear lens covers prior to stowage. When projection lens assembly is furnished mounted on projector, cover the assembly with front lens cover only. Close and adequately secure projector case cover(s).

5.1.1.5 Unit packing. Unit packing shall be in accordance with the methods and submethods prescribed in MIL-P-116 as specified herein, insuring compliance with the applicable requirements of that specification.

5.1.1.5.1 Technical literature. Each technical literature shall be individually unit packed submethod IC-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

5.1.1.5.2 Projection set, motion picture sound without transit-storage case. Each set shall be individually unit packed submethod IA-14 as follows: Cushion each secured projector on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class domestic, designed to protect the item and packaging components from physical damage during handling, transit and storage. As an alternate, cushioning material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514 or any other cushioning material complying with federal or military specifications which will provide the protection specified. Inclose the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class domestic. Close the box in accordance with the appendix of the box specification and place within a heat sealed bag conforming to MIL-E-6060. The bagged item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements table. Place the technical literature, unit packed as specified in 5.1.1.5.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.1.5.3 Projection set, motion picture sound in transit-storage case. Place the projection set prepackaged as specified in 5.1.1.4 within the transit-storage case and adequately secure fasteners. The secured transit-storage case with contents shall be individually unit packed as specified in 5.1.1.5.2.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1, except that when the projection set is to be furnished in a transit-storage case method III shall be substituted for submethod IA-14, the interior domestic fiberboard box and heat sealed bag shall be omitted and cushioning material shall be weather-resistant.

5.2 Packing. Packing shall be level A or B as specified (see 6.2b). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents when tested in accordance with methods 5016 and 5017 of FED-STD-101.

5.2.1 Level A. A multiple quantity of projection set, motion picture sound, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds (90.8 kilograms), or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds (45.4 kilograms), 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A. When a single unit is shipped to a single destination the unit shall be individually packed as specified above.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (113 kilograms) (excluding the pallet) or a volume of 20 cubic feet (0.56 cubic meters); and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

5.2.2 Level B. A multiple quantity of projection set, motion picture sound, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds (113.5 kilograms). Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the appplicable box specification. When the gross weight exceeds 200 pounds (90.8 kilograms), or the container length and width is 48 x 24 inches (1.22 x 0.61 meters) or more and the weight exceeds 100 pounds (45.4 kilograms), containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (113.5 kilograms) (excluding the pallet) or a volume of 20 cubic feet (0.56 cubic meters); and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of projection set, motion picture sound, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The motion-picture projection set is a special purpose portable commercial set with minimum modification, if necessary, to meet requirements of this specification. It is used to project 16mm optical sound film under field conditions (see 1.1).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A or level B preservation and packing (see section 5).

c. Number of first article sample units to be submitted for approval (see 3.2).

d. Place of final inspection.

e. Submission of statement of conformance referenced in 3.3, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

f. When test results of first article inspection of packaging are required.

g. When the first article packaging inspection is not required.

6.3 Group C failures. Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.4 Nomenclature. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Packaging design illustrations. When specified in the contract or order, illustrations and bill of materials delineating the unit pack design that passed the first article inspection will be prepared and furnished in accordance with the provisions of Government Drawing SC-A-362100 (FSCM 80063).

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army - ER

Preparing activity:
Army - ER

(Project 6730-A113)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-55355A(ER) PROJECTION SET, MOTION PICTURE, SOUND 16MM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Electronics Research &
Development Command
ATTN: DELCS-PE
Fort Monmouth, New Jersey 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:16Z
Check the source to verify that this is the currenversion before use.